Respiration and gas exchange

Mark scheme and worked solutions for 25 questions: 8 multiple choice, 17 structured.

1 · Part 3

Paper 1 & 2 Respiration [1]

Answer: B

Worked solution

Think of ATP as the cell's "energy currency". Glucose is the fuel; respiration breaks it down and the energy released is used to make ATP. Whenever a cell needs energy (for example, for active transport or making proteins), it uses ATP. Both aerobic and anaerobic respiration make ATP.

2 · Part 3

Paper 1 & 2 Respiration [1]

Answer: D

Worked solution

Recall the word equation for anaerobic respiration in animals: glucose → lactic acid. There is no oxygen on the left and no carbon dioxide on the right. Because glucose is only partly broken down, much less energy is released than in aerobic respiration. Only row D matches all three facts.

3 · Part 3

Paper 1 & 2 Respiration [1]

Answer: A

Explanation

The germinating seeds are alive and respiring quickly. Respiration releases energy, and some of it is lost as heat, which the vacuum flask keeps in, so the temperature rises. Option B is wrong because heat cannot make the flask warmer than its surroundings by flowing in from the cooler cupboard, and the vacuum flask reduces heat transfer anyway. Option C is wrong because the flask was in the dark, so no photosynthesis took place (and photosynthesis does not release heat). Option D is wrong because evaporation cools things down; it does not release heat.

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4 · Part 3 Paper 1 & 2 Gas exchange [1]

Answer: C

Explanation

Carbon monoxide combines with haemoglobin in red blood cells, so less oxygen can be carried by the blood. Option A, carbon dioxide, is carried mainly in the plasma and does not reduce oxygen transport in this way. Option B, nicotine, is the addictive drug in smoke; it affects the heart rate and blood vessels, not haemoglobin. Option D, tar, coats the airways and contains substances that cause cancer, but it does not combine with haemoglobin.

5 · Part 3 Paper 1 & 2 Gas exchange [1]

Answer: A

Worked solution

Breathing in: muscles contract, volume up, pressure down, air in. Breathing out at rest is the reverse: muscles relax, volume down, pressure up, air out. Only row A has both muscles relaxing and the pressure increasing.

6 · Part 3 Paper 2 only Gas exchange [1]

Answer: D

Worked solution

Gas exchange in a leaf happens by diffusion. The spongy mesophyll has many air spaces, so gases can spread quickly through the inside of the leaf to every cell. Photosynthesising cells use carbon dioxide, keeping its concentration low inside the cells, so carbon dioxide keeps diffusing in.

7 · Part 3 Paper 2 only Gas exchange [1]

Answer: C

Worked solution

Ask which way the concentration gradient points. In bright light, photosynthesis adds oxygen to the air spaces faster than respiration removes it, so there is more oxygen inside the leaf than outside. Oxygen therefore diffuses out through the stomata.

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8 · Part 3

Paper 2 only | Gas exchange | [1]

Answer: B

Worked solution

Use the colour key: more carbon dioxide gives yellow, less gives purple, no change stays red. Then decide what the leaf does in each condition. Dark: respiration only, carbon dioxide released, yellow. Bright light: photosynthesis faster than respiration, carbon dioxide taken in, purple. No leaf: no change, red.

9 · Part 3

Paper 1 & 2 | Respiration | [6]

Mark scheme

(a) **[2]**

- glucose + oxygen on the left (reactants) (1)
- → carbon dioxide + water on the right (products) (1)

ALLOW energy or ATP shown as a product

REJECT any equation that includes light or chlorophyll for the second mark

(b) **[2]**

- correct formulae: $C_6H_{12}O_6 + O_2 \rightarrow CO_2 + H_2O$ (1)
- correctly balanced: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ (1)

ALLOW subscripts written on the line, e.g. C6H12O6

IGNORE energy or ATP shown as a product

(c) **[2]**

- no energy is released or no ATP is produced (1)
- the cell processes that need energy cannot take place, e.g. active transport, making proteins, cell division, growth (1)

Worked solution

(a) Aerobic respiration uses glucose and oxygen and produces carbon dioxide and water: glucose + oxygen → carbon dioxide + water.

(b) Glucose is $C_6H_{12}O_6$. Each glucose molecule needs six oxygen molecules and produces six carbon dioxide molecules and six water molecules: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$. Check: 6 carbon atoms, 12 hydrogen atoms and 18 oxygen atoms on each side.

(c) Respiration is how cells release energy from glucose to make ATP, and ATP provides the energy for all the work a cell does. If respiration stops, no more ATP is made, so processes such as active transport, building proteins and cell division stop, and the cell dies.

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10 · Part 3 Paper 1 & 2 Respiration [5]

Mark scheme

(a) **[1]**

- any one from:
- anaerobic respiration does not need oxygen (but aerobic respiration does) (1)
- anaerobic respiration releases less energy OR makes less ATP from each glucose molecule (1)

IGNORE references to lactic acid, carbon dioxide or water (excluded by the question)

(b) **[2]**

- at low speeds (up to 12 km per hour) the concentration stays low or rises only slightly (1)
- at higher speeds (above 12 km per hour) the concentration increases rapidly or steeply (1)

ALLOW one mark for "as speed increases, lactic acid concentration increases" if neither point is made

ALLOW correct use of data for either point, e.g. 1.0 to 1.4 mmol per dm^3 from 8 to 12 km per hour; 1.4 to 8.6 mmol per dm^3 from 12 to 18 km per hour

(c) **[2]**

- at high speeds the muscles need more energy than aerobic respiration can supply OR oxygen cannot be supplied to the muscles fast enough (1)
- so the muscle cells also respire anaerobically, which produces lactic acid (1)

Worked solution

(a) Aerobic respiration needs oxygen; anaerobic respiration does not. Anaerobic respiration also releases much less energy (makes much less ATP) from each glucose molecule, because the glucose is only partly broken down.

(b) Look at how the numbers change. From 8 to 12 km per hour, lactic acid only rises from 1.0 to 1.4 mmol per dm^3. From 12 to 18 km per hour, it rises from 1.4 to 8.6 mmol per dm^3, which is a much steeper increase.

(c) At low speeds the heart and lungs supply enough oxygen for the muscles to release all the energy they need by aerobic respiration. At high speeds the muscles need energy faster than oxygen can be delivered, so some glucose is respired anaerobically, and this makes lactic acid, which passes into the blood.

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11 · Part 3 Paper 1 & 2 Respiration [5]

Mark scheme

(a) **[2]**

- glucose → ethanol (1)
- + carbon dioxide (as a second product) (1)

ALLOW alcohol for ethanol

ALLOW energy or ATP shown as a product

REJECT lactic acid; REJECT oxygen on either side

(b) **[3]**

- active transport needs energy OR needs ATP (from respiration) (1)
- anaerobic respiration releases less energy OR makes less ATP from each glucose molecule than aerobic respiration (1)
- so less energy or ATP is available for active transport (so fewer ions are taken up per unit time) (1)

ALLOW root cells cannot respire aerobically because there is little oxygen, as an alternative for the first mark

IGNORE restating that the root cells respire anaerobically (given in the question)

Worked solution

(a) In plant cells, anaerobic respiration breaks glucose down into ethanol and carbon dioxide: glucose → ethanol + carbon dioxide.

(b) Waterlogged soil has very little oxygen, so the root cells have to respire anaerobically. Anaerobic respiration only partly breaks down glucose, so it releases much less energy and makes much less ATP than aerobic respiration. Active transport moves mineral ions into the root cells against a concentration gradient and needs energy from ATP. With less ATP available, active transport slows down, so the plant takes up mineral ions more slowly.

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12 · Part 3

Paper 1 & 2 Respiration [7]

Mark scheme

(a) **[2]**

- it is a control OR to compare with flask X (1)
- boiled peas are dead and do not respire, so it shows that the temperature rise in X is caused by respiration in the living peas (1)

(b) **[2]**

- to kill micro-organisms (bacteria or fungi) on the surface of the peas (1)
- micro-organisms would also respire and release heat, so the temperature rise would not be caused only by the peas OR the temperature in Y would rise (1)

(c) **[2]**

- 28.4 − 20.0 = 8.4 °C OR 8.4 ÷ 4 (1)
- 2.1 °C per day (1)
- no unit or wrong unit: maximum one mark

(d) **[1]**

- vacuum flasks reduce heat loss to the surroundings (so the temperature rise can be measured) (1)

ALLOW they insulate the peas

Worked solution

(a) A control lets you check that the result is caused by the factor you are interested in. Boiling kills the peas, so they cannot respire. The temperature in Y did not change, so the rise in X must be caused by respiration in the living peas.

(b) Bacteria and fungi live on the surface of seeds. They respire too, and they also release heat. Disinfectant kills them, so any heat measured comes from the peas. It also stops the dead peas in Y from rotting and warming up.

(c) Temperature increase = 28.4 − 20.0 = 8.4 °C. Mean rate = 8.4 ÷ 4 = 2.1 °C per day. Check: adding the daily rises 2.6 + 2.2 + 2.0 + 1.6 gives 8.4 °C, and 8.4 ÷ 4 = 2.1.

(d) Peas only release a small amount of heat. An open beaker would lose this heat to the air, so the temperature would hardly change. A vacuum flask is insulated, so the heat stays inside and the rise can be measured.

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13 · Part 3 Paper 1 & 2 Respiration [6]

Mark scheme

(a) **[1]**

- temperature (1)

(b) **[2]**

- any two from, one mark each:
- number of woodlice OR mass of woodlice OR size of woodlice (1)
- volume of indicator (1)
- concentration of indicator OR starting colour of indicator (1)
- size of boiling tube OR volume of air in the tube (1)
- time the woodlice are left in the water bath before timing starts (1)

IGNORE temperature (it is the independent variable)

IGNORE "amount" of indicator without volume

(c) **[3]**

- the woodlice respire and release carbon dioxide (1)
- the carbon dioxide concentration in the tube increases, which turns the indicator yellow (1)
- at higher temperatures respiration is faster (because the enzymes that control respiration work faster), so carbon dioxide is released faster (1)

ALLOW enzymes have more kinetic energy OR more collisions between enzymes and substrate for the third mark if linked to faster respiration

Worked solution

(a) The independent variable is the one Amara changed on purpose: the temperature of the water bath.

(b) To make it a fair test, everything else that could affect how quickly the indicator changes must stay the same, for example the number of woodlice and the volume of indicator.

(c) Woodlice respire aerobically, producing carbon dioxide, which diffuses into the indicator. As the carbon dioxide concentration rises, the indicator changes from red to yellow. Respiration is controlled by enzymes, and in this range of temperatures they work faster as it gets warmer. So at 30 °C the woodlice respire faster, release carbon dioxide faster, and the indicator turns yellow in 7 minutes instead of 20 minutes at 15 °C.

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14 · Part 3

Paper 1 & 2 Respiration [6]

Mark scheme

(a) **[2]**

- (1.2 ÷ 16.0) × 100 (1)
- 7.5 % (1)

ALLOW 7.5 for two marks

(b) **[2]**

- in anaerobic respiration, glucose is only partly broken down OR is not broken down completely to carbon dioxide and water (1)
- lactic acid still contains (chemical) energy (1)

(c) **[2]**

- anaerobic respiration releases less energy or makes less ATP from each gram of glucose (1)
- so less energy is available for growth OR cell division OR making new cell materials such as proteins (1)

Worked solution

(a) Percentage = (1.2 ÷ 16.0) × 100 = 0.075 × 100 = 7.5 %. Check: 7.5 % of 16.0 kJ is 0.075 × 16.0 = 1.2 kJ.

(b) Aerobic respiration breaks glucose down completely into carbon dioxide and water, releasing almost all of its energy. Anaerobic respiration in muscles only breaks glucose down as far as lactic acid. Lactic acid still has a lot of chemical energy stored in it, so much less energy is released.

(c) The data show that anaerobic respiration releases far less energy from the same mass of glucose. Bacteria need energy (ATP) to make new cell materials and divide. Bacteria that can only respire anaerobically get less energy from the same glucose, so they make new cells more slowly.

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15 · Part 3

Paper 1 & 2 | Gas exchange | [6]

Mark scheme

(a) **[3]**

- the trachea divides into two bronchi, one to each lung (1)
- each bronchus divides into many (smaller) bronchioles (1)
- the bronchioles end in alveoli (1)

ALLOW trachea → bronchi → bronchioles → alveoli in the correct order for all three marks

- if structures are in the wrong order, maximum one mark

(b) **[1]**

- ribs (1)

ALLOW rib cage

(c) **[2]**

- one pleural membrane covers the surface of each lung (1)
- the other lines the inside of the thorax (chest wall) OR lines the ribs and intercostal muscles (1)

ALLOW pleural fluid between the membranes for one mark if only one position is given

ALLOW "surround the lungs" for one mark only

Worked solution

(a) Air passes down the trachea, which divides into two bronchi, one going to each lung. Inside each lung the bronchus branches many times into narrower tubes called bronchioles. The bronchioles end in tiny air sacs called alveoli, where gas exchange happens.

(b) The ribs form a cage around the lungs and protect them.

(c) There are two pleural membranes. One covers the outside of each lung, and the other lines the inside of the thorax wall. Between them is a thin layer of fluid.

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16 · Part 3 Paper 1 & 2 Gas exchange [6]

Mark scheme

(a) **[1]**

- lungs (1)

ALLOW alveoli

(b) **[3]**

- pulling the sheet down increases the volume inside the bottle (1)
- so the pressure inside the bottle decreases (1)
- air is pushed in from outside, where the pressure is higher, through the tube into the balloons OR air moves into the balloons down a pressure gradient (1)

(c) **[2]**

- any two from, one mark each:
- the bottle wall cannot move OR the model has no ribs or intercostal muscles to change the volume (1)
- the real diaphragm is moved by its own muscle contracting, not pulled by hand (1)
- the real diaphragm is dome-shaped and flattens when it contracts (1)
- in the body, the lungs fill the thorax, with no large air space around them (1)
- the model has no pleural membranes or pleural fluid (1)
- real lungs contain bronchioles and alveoli, not one hollow space like a balloon (1)

Worked solution

(a) The balloons fill with air when the volume of the bottle increases, just as the lungs fill with air when the volume of the thorax increases.

(b) Pulling the rubber sheet down is like the diaphragm contracting and flattening. It makes the space inside the bottle larger, so the air pressure inside the bottle falls. Air outside is now at a higher pressure, so it moves in through the tube and inflates the balloons. This is the same reason air enters the lungs when you breathe in.

(c) A model is never exactly like the real thing. The most important difference is that the walls of the real thorax move: the intercostal muscles move the ribs, which also changes the volume of the thorax. The plastic bottle cannot do this. Also, the real diaphragm is a muscle that contracts by itself, and the lungs fill the thorax rather than hanging in a large air space.

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17 · Part 3

Paper 1 & 2 | Gas exchange | [6]

Mark scheme

(a) **[4]**

- indicative points, any four from, one mark each:
- there are very many alveoli, giving a large surface area for diffusion (1)
- the alveolus wall is very thin (one cell thick), giving a short diffusion distance (1)
- capillary walls are also thin OR capillaries are very close to the alveolus wall, so the diffusion distance is short (1)
- each alveolus is surrounded by many capillaries OR a good blood supply, which carries oxygen away and brings carbon dioxide, keeping a steep concentration gradient (1)
- ventilation (breathing) brings in fresh air and removes stale air, keeping a steep concentration gradient (1)
- the inner surface is moist, so gases dissolve before diffusing (1)
- a feature with no explanation of how it helps diffusion: maximum two marks

(b) **[2]**

- 70 ÷ 1.75 (1)
- 40 (times) (1)

Worked solution

(a) Diffusion is faster when the surface area is large, the distance is short and the concentration gradient is steep. Alveoli have all three. Millions of alveoli give a huge surface area. The alveolus wall and the capillary wall are each only one cell thick, so oxygen and carbon dioxide only have a very short distance to travel. Blood flowing through the capillaries keeps taking oxygen away and bringing carbon dioxide, and breathing keeps replacing the air in the alveoli, so the concentration gradients stay steep. The moist lining lets the gases dissolve so they can diffuse through the cells.

(b) 70 ÷ 1.75 = 40, so the surface area of the alveoli is about 40 times larger than the surface area of the skin. Check: 1.75 × 40 = 70.

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18 · Part 3 Paper 1 & 2 Gas exchange [7]

Mark scheme

(a) **[2]**

- the more cigarettes smoked each day, the higher the death rate from coronary heart disease (1)
- correct use of data, e.g. men who smoked 25 or more a day had three times the death rate of non-smokers OR the death rate rose from 160 to 480 per 100 000 men per year (1)

(b) **[2]**

- (480 − 160) ÷ 160 × 100 OR 320 ÷ 160 × 100 (1)
- 200 % (1)

ALLOW 200 for two marks

(c) **[3]**

- any three from, one mark each:
- chemicals in smoke (e.g. nicotine or carbon monoxide) damage the lining of the arteries (1)
- nicotine increases heart rate OR increases blood pressure (1)
- fatty deposits build up in the walls of the coronary arteries (1)
- the coronary arteries become narrower OR a blood clot can form and block them (1)
- less blood, so less oxygen, reaches the heart muscle (1)
- heart muscle cells cannot respire aerobically enough and may die OR this can cause a heart attack (1)
- carbon monoxide reduces the oxygen carried by the blood, so the heart has to work harder (1)

Worked solution

(a) As the number of cigarettes smoked each day increases, the death rate from coronary heart disease increases. Non-smokers had 160 deaths per 100 000 men per year, but men smoking 25 or more a day had 480, three times as many.

(b) Increase = 480 − 160 = 320. Percentage increase = (320 ÷ 160) × 100 = 200 %. Check: a 200 % increase means adding twice the original value: 160 + 320 = 480.

(c) Coronary arteries supply the heart muscle with blood. Chemicals in tobacco smoke damage the artery lining and nicotine raises the blood pressure, so fatty deposits build up in the artery walls. The coronary arteries become narrower and a blood clot may block them. The heart muscle then receives less oxygen, cannot respire aerobically enough to release the energy it needs, and part of it may die. This is a heart attack.

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19 · Part 3 Paper 1 & 2 Gas exchange [6]

Mark scheme

(a) **[3]**

- the cilia stop beating or are destroyed (1)
- so mucus, with trapped dust and micro-organisms, is not moved away from the lungs OR mucus builds up in the airways (1)
- coughing is the only way to remove the mucus (from the airways) (1)

ALLOW smoke irritates the airways so more mucus is made, for the second mark

(b) **[3]**

- the alveoli have a smaller surface area (1)
- so less oxygen diffuses into the blood OR gas exchange is slower (1)
- less oxygen reaches the muscles for aerobic respiration OR carbon dioxide is not removed fast enough, so the person breathes faster or deeper to try to get enough oxygen (1)

Worked solution

(a) Normally the cilia sweep mucus, with the dust and micro-organisms it has trapped, up out of the airways so it can be swallowed. When the cilia are damaged, the mucus stays in the trachea and bronchi and builds up. The smoker has to cough to bring it up. The trapped micro-organisms also make lung infections more likely.

(b) When the walls between alveoli break down, many small alveoli become a few large air spaces. This greatly reduces the surface area for gas exchange, so less oxygen diffuses into the blood each minute. During exercise the muscles need more oxygen for aerobic respiration. The person cannot get enough oxygen into the blood, so they breathe faster and feel breathless.

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20 · Part 3 Paper 1 & 2 Gas exchange [8]

Mark scheme

(a) **[2]**

- 26 + 29 + 26 = 81 OR 81 ÷ 3 (1)
- 27 breaths per minute (1)

ALLOW 27 for two marks

(b) **[3]**

- any three from, one mark each:
- muscles contract more, so they need more energy (1)
- the muscles respire faster (aerobically) OR need more oxygen (for aerobic respiration) (1)
- more carbon dioxide is produced (by respiration) (1)
- breathing faster takes in more oxygen OR removes more carbon dioxide from the blood (1)

(c) **[1]**

- so his breathing rate had returned to its normal resting level OR so that earlier activity did not affect the resting value (1)

(d) **[2]**

- breathe out through one tube of limewater and breathe in through (or draw air in through) a second tube of limewater (1)

ALLOW bubble exhaled air through limewater and pump normal air through another tube of limewater

- the limewater that exhaled air passes through turns cloudy faster or more than the other (1)

Worked solution

(a) Mean = (26 + 29 + 26) ÷ 3 = 81 ÷ 3 = 27 breaths per minute. Check: 27 lies between the lowest (26) and highest (29) values.

(b) During step-ups, the leg muscles contract more often and need more energy. They get this by respiring faster, which uses more oxygen and produces more carbon dioxide. Breathing faster increases the gas exchange in the lungs, so more oxygen gets into the blood and more carbon dioxide is removed.

(c) If Tomas had just been walking around, his breathing rate would still be raised. Sitting still for five minutes makes sure the resting value really is his resting rate.

(d) Use two tubes of limewater. Breathe out gently through a tube into one so exhaled air bubbles through it, and breathe in through a tube so that air from the room is drawn through the other. The limewater that exhaled air passes through goes cloudy much sooner, showing that exhaled air contains more carbon dioxide.

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21 · Part 3 Paper 2 only Gas exchange [6]

Mark scheme

(a) **[4]**

- indicative points, any four from, one mark each:
- the leaf is thin, so the diffusion distance from the air to the cells is short (1)
- the leaf is broad or flat, giving a large surface area for gas exchange (1)
- (many) stomata allow carbon dioxide and oxygen to diffuse into and out of the leaf (1)
- air spaces in the spongy mesophyll allow gases to diffuse quickly to and from all the cells (1)
- mesophyll cells have a large total surface area in contact with the air spaces (1)
- the surfaces of the mesophyll cells are moist, so gases dissolve before diffusing into or out of the cells (1)
- a feature with no explanation of how it helps gas exchange: maximum two marks

(b) **[2]**

- in bright light, photosynthesis is faster, so carbon dioxide is used up faster by the mesophyll cells (1)
- so the carbon dioxide concentration inside the leaf falls lower (compared with the air outside) (1)

> **ALLOW** in bright light, photosynthesis is faster than respiration, so there is a net uptake of carbon dioxide, for the first mark

Worked solution

(a) A leaf has to get carbon dioxide to every photosynthesising cell and remove oxygen. It is thin, so no cell is far from the air, and broad, so it has a large surface. Stomata, mostly in the lower surface, are pores that let gases in and out. Inside, the spongy mesophyll has many air spaces, so gases can diffuse through the leaf quickly, and the mesophyll cells have a large, moist surface area where gases dissolve and diffuse into and out of the cells.

(b) The concentration in the air outside stays about the same, so the gradient depends on how low the concentration inside the leaf is. In bright light the cells photosynthesise faster and use up carbon dioxide faster than respiration replaces it, so the concentration inside the leaf falls further below that in the air. The gradient is steeper, so carbon dioxide diffuses in faster.

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22 · Part 3

Paper 2 only | Gas exchange | [6]

Mark scheme

(a) **[2]**

- stomata are pores in the surface of the leaf (1)
- that allow carbon dioxide and oxygen to diffuse into and out of the leaf (1)

ALLOW guard cells open and close the stomata to control gas exchange, for one mark

(b) **[2]**

- 270 ÷ (30 + 270) × 100 OR 270 ÷ 300 × 100 (1)
- 90 % (1)

ALLOW 90 for two marks

(c) **[2]**

- the lower surface is in contact with water, so gases cannot be exchanged with the air there (1)
- the upper surface is in contact with the air, which is where the leaf gets carbon dioxide from and releases oxygen to OR gases diffuse much faster through air than through water (1)

ALLOW stomata on the lower surface would let water into the air spaces, for one mark

Worked solution

(a) Stomata are small pores, mostly in the lower surface of a leaf. Carbon dioxide and oxygen diffuse into and out of the leaf through them.

(b) Total stomata = 30 + 270 = 300 per mm^2. Percentage on the lower surface = (270 ÷ 300) × 100 = 90 %. Check: 10 % of 300 is 30, the number on the upper surface.

(c) Gas exchange with the air is only possible where the leaf touches the air. The lower surface of the floating leaf is in the water, so stomata there would be useless and might let water flood the air spaces. The upper surface faces the air, so the stomata there let carbon dioxide in and oxygen out.

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23 · Part 3 Paper 2 only Gas exchange [7]

Mark scheme

(a) **[1]**

- 10 (arbitrary units) (1)

(b) **[2]**

- the rate of photosynthesis equals the rate of respiration (1)
- so the carbon dioxide used in photosynthesis equals the carbon dioxide released by respiration (1)

(c) **[2]**

- there is no light, so no photosynthesis takes place (1)
- respiration continues and releases carbon dioxide, so there is a net release of carbon dioxide OR the plant releases carbon dioxide at a rate of 4 units (1)

(d) **[2]**

- rate of respiration = 4 (arbitrary units) OR 10 + 4 (1)
- 14 (arbitrary units) (1)

ALLOW ECF from an incorrect respiration rate stated in the working

Worked solution

(a) The net rate of carbon dioxide uptake is 0 at a light intensity of 10 arbitrary units.

(b) Respiration releases carbon dioxide and photosynthesis uses it. At this light intensity the two processes happen at the same rate, so all the carbon dioxide released by respiration is used in photosynthesis, and none is taken in from or given out to the chamber.

(c) In the dark there is no photosynthesis, but the plant still respires. Respiration releases carbon dioxide, so the net rate of uptake is negative (−4), meaning 4 units are released.

(d) At a light intensity of 0 only respiration happens, so the rate of respiration is 4 units. If this rate stays the same, then at a light intensity of 40: rate of carbon dioxide use in photosynthesis = 10 + 4 = 14 arbitrary units. Check: 14 used − 4 released = 10 net uptake, which matches the table.

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24 · Part 3 Paper 2 only Gas exchange [7]

Mark scheme

(a) **[1]**

- the colour of the indicator OR the carbon dioxide concentration in the tube (1)

(b) **[3]**

- the light intensity is high (1)
- so the rate of photosynthesis is greater than the rate of respiration (1)
- so there is a net uptake of carbon dioxide by the leaf, the carbon dioxide concentration in the tube falls, and the indicator turns purple (1)

(c) **[2]**

- heat raises the temperature, which changes the rate of photosynthesis or respiration (because enzymes work at different rates) (1)
- place a beaker or tank of water between the lamp and the tubes OR use an LED lamp OR place the tubes in a water bath at a constant temperature (1)

(d) **[1]**

- to show that the colour change is caused by the leaf OR that light (or heat) alone does not change the colour of the indicator (1)

Worked solution

(a) The dependent variable is what Femi observed to measure the effect: the colour of the indicator, which shows the carbon dioxide concentration.

(b) At 10 cm the light is brightest. The leaf photosynthesises faster than it respires, so it takes more carbon dioxide out of the air in the tube than it adds. The carbon dioxide concentration falls below that of normal air, so the indicator changes from red to purple.

(c) Tubes nearer the lamp would be warmer as well as brighter, so temperature becomes a second variable that could change the rates of photosynthesis and respiration. A heat shield such as a tank of water, an LED lamp, or a water bath keeps the temperature the same for every tube.

(d) The tube with no leaf stayed red, which shows that the indicator does not change colour just because it is near the lamp. Any colour change in the other tubes must be caused by the leaves.

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25 · Part 3 Paper 2 only Gas exchange [6]

Mark scheme

(a) **[2]**

- the concentration increased from 06:00 to a maximum of 9.4 mg per dm^3 at 18:00 (1)
- it then decreased through the night, back to 4.2 mg per dm^3 at 06:00 the next day (1)

ALLOW "increases during the day and decreases during the night" for one mark if no data are used

(b) **[4]**

- indicative points, any four from, one mark each:
- during the day the plants photosynthesise, releasing oxygen (1)
- during the day the rate of photosynthesis is greater than the rate of respiration, so there is a net release of oxygen into the water (1)
- the oxygen diffuses out of the plants into the water (1)
- the concentration is highest at 18:00 because oxygen has been added throughout the hours of daylight (1)
- at night there is no light, so no photosynthesis takes place (1)
- plants and animals continue to respire day and night, using oxygen, so the concentration falls at night (1)
- the concentration is lowest at dawn (06:00) because oxygen has been used all night without being replaced (1)

Worked solution

(a) The dissolved oxygen concentration rose from 4.2 mg per dm^3 at sunrise (06:00) to a peak of 9.4 mg per dm^3 at sunset (18:00). It then fell all night, back to 4.2 mg per dm^3 at 06:00 the next day.

(b) In daylight the water plants photosynthesise. In the light, photosynthesis is faster than respiration, so the plants release more oxygen than they use, and the extra oxygen diffuses into the water. Oxygen is added all day, so the concentration is highest at sunset. After dark there is no photosynthesis, but the plants and the animals in the pond keep respiring, which uses up oxygen. With nothing replacing it, the concentration falls all night and is lowest just before sunrise.

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